

Chapter 5

Desertification Control through Floodwater Harvesting: The Current State of Know-How

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Abstract The ecologically disastrous sedentarization of nomads in the Gareh Bygone Plain, southern Iran, along with the application of inappropriate technologies, ploughs and pumps, has desertified a sandy scrubland teeming with wildlife. This resulted in *mass* cityward migration of the unskilled nomads. Floodwater harvesting for the artificial recharge of groundwater has reversed the migration tide and converted consumers into producers again. We have proven that floodwater harvesting can mitigate flooding and drought, and modify the marginalities of climate and soil for a sustainable agricultural production. We have harvested upwards of 200 million cubic meters of floodwater since 1983, of which 80% has recharged the aquifers. A few of our findings are:

- The slope of the conveyor-spreader channel is 0.0003 along 85% of its length, decreasing to zero for the next 5%, and flat for the last section.
- The safe distance between the channels on low slopes is 70–100 m.
- The number of gaps and their capacity should increase in the downstream direction as a safeguard in case of breaching of embankments during exceptional floods.
- The mean annual yield of floodwater-irrigated indigenous range plants is 445 kg per ha, 5 fold that of the control. The quail bush planted at 625 seedlings per ha is 1,500 kg.
- The mean annual stem and fuel wood yield of river red gum is 4.68 and 0.78 t per ha. The annual mean above ground C-sequestration of same trees is 2.97 per ha. Root channels of these trees facilitate deep percolation.

Keywords Floodwater harvesting, artificial recharge of groundwater, indigenous technology

1 Introduction

Water, the most precious commodity in deserts and the substance without which life as we know it is impossible, is also a desertification agent. Not only does it degrade the land through erosion, salinization and inundation, but it also provides the material transported by the wind. Moreover, flooding is a major hazard that causes untold damage and misery. Therefore, water management is often the key to many aspects of desertification control (Mabbutt, 1987). As freshwater is one of the ecosystem services deemed ‘well beyond levels that can sustain current demands, let alone future ones’ (Anon., 2005), harvesting this irreplaceable substance and protecting it from pollution is extremely important in deserts.

A common misconception of laymen about deserts is their extreme water shortage. This wrong notion has developed through the ages from travellers who had crossed the barren land, missing the chance to be soaked by the torrential rains and observe the roaring floods. A classic case of this is the convective storms that usually occur on summer afternoons and that ruin the life and belongings of those living in deserts. Application of a time-proven indigenous technology could mitigate many of the water-related problems in deserts and break the downward spiral of factors that lead to desertification.

According to Glantz (1994) the marginalities of the climate, soil and topography are deterrents to a sustainable agricultural production. The ancient Persians, who lived on desert margins, faced the former two constraints in the plains and all three constraints in the foothills. Furthermore, as many other desert-dwelling nations, they had to contend with the extreme meteorological events of floods and droughts. Therefore, they turned the problem into a solution by developing the art of “water husbandry”, living in deserts through harvesting the ephemeral floods for runoff farming, and using it prudently. This drought-coping mechanism mitigated the flooding hazards, too. A related advantage of this simple practice, besides storing the surplus water in underground reservoirs, was building soil by depositing the nutritious sediments on severely eroded land, thus simultaneously mitigating ‘pseudo-drought’ (Wolman, 1985) and modifying the topography. The Persians regained upwards of 600,000 ha of land from deserts using the multipurpose water harvesting techniques. This paradigm-based technology led to the invention of *qanat*, the best means of water collection and delivery in hot climates where agricultural drought is an ever present threat to the very existence of desert-dwellers.

Although the application of an inappropriate technology, namely pumping of water from any imaginable depth, has made more than one-half of our qanats inoperative, the sound scientific basis – water storage in potential aquifers – is perfectly valid today. This variation on the theme of “genesis strategy” is most appropriate where potential, easy to reach aquifers exist.

This principle is implemented for recharging the overexploited and potential aquifers, along with other benefits accrued from harvesting the turbid floodwater, such as desertification control (Kowsar, 1991). In a world facing the diminishing availability of arable land, this is of vital importance, as most of the non-marginal

land has already been taken. Therefore, we have to build arable land to ensure that future generations have the right to life, too!

Recurrent and prolonged droughts, exacerbated by overpopulation and overdevelopment of a fragile ecological landscape, desertified the Gareh Bygone Plain, which was once a healthy scrubland teeming with wildlife. The elders, who massacred the gazelles, reported encountering flocks of over 100 of them 50 years ago. The advent of diesel pumps and moldboard ploughs hitched to powerful tractors delivered the *coup de grâce* to this productive land. The plain was reminiscent of a moonscape when we first arrived there on 6 November 1982. The depth of the water table, which had been 10 m before the arrival of the first pump, was 26 m in a well dug in the summer of 1982. This well went dry after only 10 min of pumping by a 24 hp diesel engine through a 4 in. (10.6 cm) pipe. *Mass* cityward migration of mixed farmers was a consequence of anthropological desertification.

2 Floodwater Harvesting Concepts

1. While the mean annual rainfall in drylands is very low, its variability is extremely high (erratic in distribution and frequency). The chance of receiving the desired rainfall at the expected time is meagre, indeed. Therefore, water is the most precious commodity, and flood is its largest supplier in drylands. Although floodwater is a renewable capital, its use must be optimized.
2. Available water capacity (AWC) is the most important direct driver of ecological sustainability in drylands. Soil *texture* and *depth* are the two major determinants of AWC.
3. Water, an erosion agent, may become a land renovator and soil builder by transporting to and depositing nutritious sediment on the slightly sloping, drastically disturbed lands, thus improving their texture and increasing their depth, and enhancing their AWC. This process also modifies the topography, stabilizes the moving sands and deprives the wind of the erodible material at its source.
4. High evaporation rate from surface waters, rapid siltation, inundation of productive lands and dwellings, forced migration of the inhabitants of the inundated area and construction sites, threat to biodiversity, reservoir leakage, earthquakes and other related environmental hazards, and the very high costs and relatively long time needed for their construction, make large dams the most inappropriate technology in drylands. Furthermore, such schemes do not benefit the inhabitants of the runoff-producing headwater catchments. They not only lose most of the water that nature bestows upon them, but also the surface soil, the life-giving substance that nothing can replace.
5. Turbid floodwater should be harnessed to build soil and produce virtual water through spate irrigation, and/or it should be stored in aquifers by employing artificial recharge of groundwater (ARG) methods and used commensurate with needs. Doing so would help to mitigate flooding hazards.

3 Materials and Methods

Soil and water quality and the presence or absence of potential aquifers are the determining factors in planning and implementing optimization of their use. Three methods may be used:

1. **Freshwater and clean alluvium.** This is the simplest form of ARG. Runoff from rocky outcrops (massifs, batholiths, some conglomerates, etc.), which is devoid of eroded material or snowmelt water of low velocity, is suitable for recharging clean alluvium and karst formations. The very high infiltration rate of these materials necessitates construction of the smallest recharge ponds. Infiltration rate of the Ab Barik, Bam ARG system in southeastern Iran, exceeds 10 m per day; i.e., each square meter of a recharge pond floor could pass 10 m³ of freshwater in 24 h. The extent, depth and macro-porosity of the alluvium determine the volume of the underground reservoir. Although the ARG of karst formations and confined aquifers is practical when employing this method, the necessity of detailed, and very expensive, geological studies eliminates this option for most deserts.
2. **Spate irrigation of agronomic crops and adopted trees.** Small-grained cereals (e.g., wheat and barley in cold and temperate climates, sorghum and millet in temperate and hot climate) are suitable crops for runoff farming. Perennial or annual forages may be programmed into the farming plans. Adapted annual legumes (e.g., *Trifolium alexandrinum* L. in mild and hot climates) fit nicely with fallowing in cereal production. Incidentally, this legume is suppressive and checks the growth of most weeds. Establishment of woodlots and growing fruit trees is another possibility where potential aquifers are nonexistent. Fodder trees and shrubs are particularly important as they provide a minimum of forage for a few livestock that may be used for building a healthy flock when the prolonged hydrological drought is over. Covered cisterns and disinfectants can provide hygienic water for domestic consumption. Rooting depth and water requirement of the intended crops, the AWC of the soil, and the estimated annual or biennial expected floodwater volume determine the area needed for its harvesting by means of the floodwater spreading (FWS) techniques. The system consists of a conveyance canal, a conveyor-spreader channel (CSC), a few level-silled channels and a tail drain. Installation of a cistern or a tank upstream of the FWS system and another one ahead of the tail drain ensures an adequate supply of domestic water. Phillips (1957), Newman (1963) and Quilty (1972a, b) have presented the methods of FWS system construction. Our contribution to the overall success of this technique is the design of the CSC and the introduction of the safe distance between the channels. These and other relevant points for the design, operation and maintenance of the FWS systems, acquired during 34 years of work throughout Iran but mostly in the Gareh Bygone Plain in the southern part of the country, will be presented later.
3. **Spate irrigation along with the ARG.** At least 6 million ha of arable soils in Iran are underlain with coarse-grained calcareous alluvium of good quality.

A large amount of sands and sandy soils have the same property. This provides an excellent opportunity for the simultaneous irrigation of crops and trees, and the ARG. Assuming identical flows in these three situations, the FWS area is larger than the first and smaller than the second system. Here we are concerned with neither the water requirements of plants nor the AWC of the soil, because the water that does not remain in the rooting zone recharges the aquifer. As the clogging of the vadose zone by the very small clay-sized particles is inevitable (Mohammadnia and Kowsar, 2003), planting of deep-rooted plants, particularly fodder trees, is strongly advised. *Acacia salicina* Lindl., which is a fodder tree and supplies nectar and pollen for honey bees from September to March in relatively warm climates, is a good candidate. Introduction of burrowing macrofauna, such as the sowbug (Kowsar, 2005) and the dung beetle, facilitates formation of preferential flow paths in the deep and hard crusts.

4 Reclamation of a Desert

The politically expedient and ecologically disastrous sedentarization of nomads in the 1930s in the Gareh Bygone Plain, southern Iran (28° 35' N, 53° 53' E, 1,140 m above mean sea level), with the mean annual precipitation of 243 mm (the coefficient of variation = 0.46) and Class A pan evaporation of 3,200 mm, wreaked havoc on a sandy scrubland teeming with wildlife. Overgrazing and fuelwood collection, the inevitable outcome of concentrating transhumant nomads in a limited area along with the application of inappropriate technologies of moldboard plows and pumps, made life unbearable for most settlers. The water table receded 10 m in 12 years, very close to the bedrock. Saltwater intrusion into the aquifer compounded the water shortage problem. Soil salinization was the outcome of irrigation with saline waters. These conditions resulted in the *mass* cityward migration of the nomads. Only a few hardy souls stayed and eked out a meagre living from the remaining natural resources.

Reversing the migration tide was the logical way to relieve the overcrowded cities of unskilled people who eke a living, sometimes by illegal means. Brainstorming sessions with the refugee elders revealed that supplying abundant and safe freshwater was the most important incentive for their return to their former homes and restarting mixed farming enterprises. Thus, the provision of water security became the litmus test of the government's integrity.

Floodwater harvesting for the artificial recharge of groundwater has reversed the migration tide and converted consumers into producers again. We have proven that floodwater harvesting can mitigate flooding and drought, and modify the marginalities of climate and soil for sustainable agricultural production.

We have harvested upwards of 200 million cubic meters of floodwater since 1983, of which over 160 million cubic meters have recharged the aquifers. Furthermore, the turbid floodwaters have deposited about 2 million tons of nutritious

suspended loads in the sedimentation basins, converting them into arable land. More than 1,000 ha of the moving sand have been fixed and become arable with the result that there are no more dust storms from the floodwater spreading areas. The area of irrigated fields has increased eight-fold to 1,193 ha. There are reportedly 200 wells, 12.5 times the number in 1983. This is bad news because we should have 60 high-yielding wells at most. Unfortunately, expediency prohibits the responsible authorities from taking harsh actions against the infringers. Our findings are as follows:

- The most frequently asked question in FWS concerns the slope of the spreading area, as it affects the erosive power of surface flows. We recommend 2% as the maximum slope of the annually ploughed land. However, we have worked on slopes of up to 6% by constructing contour furrows at suitable intervals (50–40 m for 3% and 4% slopes, and 30–20 m for 5% and 6% slopes, respectively). This spacing may be increased on non- and minimum-tilled fields. FWS on range-lands and pastures of 5–6% slope is implemented provided that they have good vegetative cover.
- To obtain a rather uniform water distribution along the head of the command area, we lay the conveyor-spreader channel at 0.0003 gradient along 85% of its length, starting at its junction with the conveyance canal. The gradient is gradually reduced to zero in the adjacent 5% of the channel's length. The final 10% is flat. This final section ensures a good harvest in years with hydrological drought, when the system does not receive the design flow (Fig. 1).
- The distance between the channels should be determined in such a way that the flow does not acquire an erosive velocity. Empirically, we have found 100 m to be a suitable distance. Performing detailed hydraulic studies, Adelpoor (personal communication) has arrived at 70 m for our conditions – loamy sand at 0.6% gradient.
- A very important point to remember is to cultivate only on the contour, leaving a 3 m wide sill of the channels out of cultivation. The gaps in the embankments should be covered with perennial grasses that are not considered weeds for the crops grown in the system. Installation of masonry chutes and drops may become relevant in very erodible soils.
- A common cause of chute collapse is the growth of tree roots below it. The root channels formed due to decomposition of the dead roots results in piping that erodes the soil beneath the chute and undermines it (Fig. 2). To prevent this loss, we select 10 m as the closest distance between a tree and the wing wall of a chute.
- The presence of geologic nitrogen in the watershed that generates flow is a financial advantage provided that its concentration in floodwater is not harmful. This may eliminate the need to apply N-fertilizers to crops. *Eucalyptus camaldulensis* Dehnh. and many other plants filter this element from percolating water. Moreover, the livestock dung carried by floodwater not only provides some of the nutrients to the vegetative cover but also increases the soil organic matter. As the vadose zone filters most of the pollutants and pathogens from the recharge water, groundwater of good quality will be available in the aquifer.

- Formation of a thick crust on the surface and deposition of very fine clay minerals in the vadose zone (Mohammadnia and Kowsar, 2003) is the natural outcome of using turbid floodwaters for the ARG. Fortunately, the sowbug (*Hemilepistus shirazi* Schuttz), which drills burrows 7 mm in diameter and up to 180 cm deep, and *Eucalyptus camaldulensis* Dehnh. trees, which take root down to the water table (30 m so far), have solved the problem. As a scarab beetle (*Peltotrupes youngi* Howden) may burrow to depths of 5 m (Kalisz and Stone, 1984), the introduction of the sowbug and deep-burrowing adapted beetles is therefore a strategy against the eventual clogging of the ARG systems. The dung beetle (*Helicopriss andersoni* Hope) is a good candidate for floodwater-irrigated pastures that serve as sedimentation basins (Fig. 3).
- Selection of trees, shrubs and bushes for enhancing the hydraulic conductivity of the vadose zone is of utmost importance. Only the thrifty, drought-enduring, nitrogen-fixing trees, shrubs and bushes (pioneer species) that do not consume a lot of water when it is available should be planted. Moreover, the number of stems per ha should be the minimum required for facilitating percolation. We have made the mistake of planting *Eucalyptus camaldulensis* Dehnh. at 3×3 m spacing in the Gareh Bygone Plain. Each single 6-year-old tree consumes about 50 l of water per day in summer, more than the per capita requirements in the same area. This tree, which can tolerate drought for a long time, evolves into an efficient pump when water is available. Each ha of this tree deprives a community with 1,000 residents of domestic water.
- The loss of water holding capacity, a consequence of soil erosion, is an important factor in the processes leading to desertification. Soil building on eroded land is a companion benefit of application of FWS methods, particularly in sandy deserts. This is of utmost importance in a world facing accelerated land degradation and loss of arable land, as well as a water-shortage crisis.
- Contrary to common knowledge, sandy soils are suitable for growing small grains and range plants if floodwater carries silt and clay particles. Moreover, we have observed grain wheat yields of up to 5 t per ha downstream of our station in a loamy sand irrigated at a three-week interval. The normal irrigation cycle on soils of finer texture in our area is 10–14 days. This anomaly requires a methodical study.
- In hot deserts embankments provide a humid and cool place for rodents and small carnivores. Flow of floodwater through their burrows usually ends up in embankment breaching. As this mishap causes a surge, and this may cause chain breaching of downstream embankments, the overall capacity of gaps for conveying the flow in each downstream embankment must therefore increase relative to its upstream one. Positioning larger gaps in strategic places overcomes this problem.
- Headwater erosion, where surplus water returns to a waterway, may end up in the irreversible destruction of the FWS systems. Positioning the outlet on exposed rocks, or constructing a solid drop or a chute with an adequate stilling basin, prevents this process.

- Simple gabion aprons, placed directly on the coarse alluvium or exposed rock riverbeds, are the most economical structures for floodwater diversion into the inlet of conveyance canals. As sedimentation in the FWS area raises its level, crown elevation of the apron should be raised by adding rocks to the gabion, or placing another layer on top of the original one. These aprons have right triangular cross-sections in which the shortest side forms the upstream wall, and the hypotenuse functions as the chute. We have successfully tried such aprons on streambeds with a slope of up to 2%.
- A solid structure at the inlet of the inundation canal prevents its enlargement during exceptionally high floods. We have observed the complete diversion of a river through an unarmored inlet.
- A very interesting discovery, which is important when working in sandy deserts, is the needleleaf sedge (*Carex stenophylla* Wahl.), a rhizomatous plant that forms a 5–10 cm armour that holds sand in place. This species is a drought-avoiding plant that inhabits hot areas with a mean annual rainfall of 150 mm. It is important not to plough through this armour if you do not have enough water to irrigate the land all year round.
- Quail bush *Atriplex lentiformis* Torr., at 625 seedlings per ha, can provide 1,500 kg of nutritious fodder for three livestock all year round. This very valuable C₄ shrub, which is the favourite food of sowbugs, can neither regenerate itself in our environment nor tolerate inundation. However, an equally palatable indigenous species but of lower yield, *Atriplex leucoclada* (Boiss.), is replacing it.
- The ten-year mean yield of native, floodwater-irrigated forage has been 445 kg per ha, five-fold the control (Fig. 4).
- The mean annual honey yield of our ARG systems, planted with eucalypts and acacias, is 200 kg per ha. We have not yet reached potential production.
- The carbon sequestration potential of the ARG systems is remarkable. Above ground, the annual C-sequestration of an 18-year-old stand of *Eucalyptus camaldulensis* Dehnh. at the Gareh Bygone Plain is 2.97 t per ha. For *Acacia salicina* Lindl. this is 1.30 t per ha per year. This indicates the potential of deserts for global warming mitigation. Signatories of the Kyoto Protocol are offered a great opportunity to green the deserts through their carbon rent contribution in the context of the Clean Development Mechanism. The mean annual stem and fuel wood yield of the eucalyptus at age 18 has been 4.68 and 0.78 t per ha, respectively (Fig. 5).

5 Conclusions

Floodwater harvesting for desertification control is a positive step towards achieving five of the eight Millennium Development Goals by 2015:

1. It provides the resources for food production and employment opportunities for those who are genuinely eager to work (Goal 1).



Fig. 1 The faint, straight line in the upper part of the picture is the conveyor-spreader channel that flows from right to left. In this event, which occurred on 6 March 1983, $5 \text{ m}^3 \text{ s}^{-1}$ of floodwater was spread along 1,000 m of this 1,342 m canal. The mean depth over the sedimentation basin was 5 cm, and the velocity was 20 cm^{-1} .



Fig. 2 Copious growth of eucalyptus roots has undermined a masonry chute. The minimum distance between a *Eucalyptus camaldulensis* Dehnh. tree and a solid chute should be 10 m.

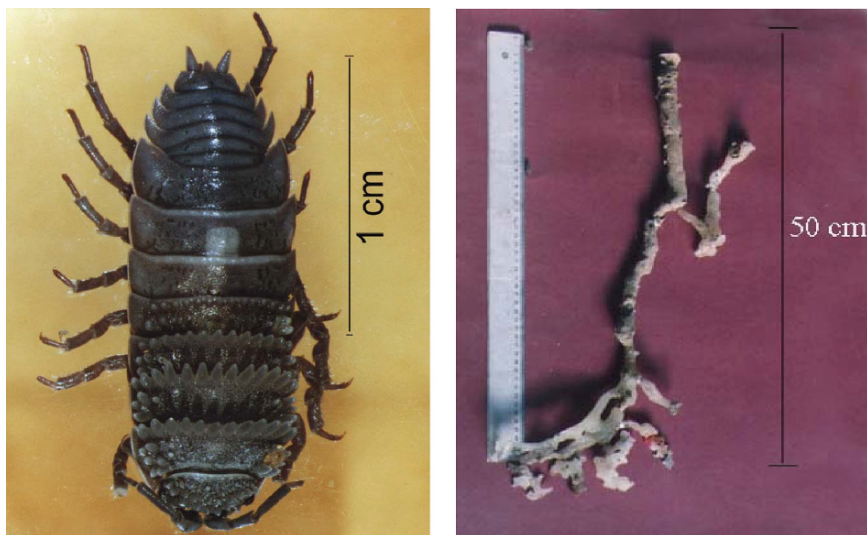


Fig. 3 The sowbug (*Hemilepistus shirazi* Schuttz) and a cast of its burrow. Were it not for this crustacean and the root channels, clogging of the vadose zone would have occurred (Gholamreza Rahbar).



Fig. 4 Floodwater irrigation of a depleted rangeland has increased its annual yield 5 times to 445 kg ha^{-1} . The mean annual yield of quail bush (*Atriplex lentiformis* (Torr.)) adds another $1,500 \text{ kg ha}^{-1}$ to the forage yield of the floodwater spreading systems. These and the fodder trees have increased the grazing capacity of the systems by about thirty-fold.



Fig. 5 The author is standing at the very place that a man was standing 22 years before this picture was taken and holding the photograph of the second floodwater spreading event. The annual above-ground carbon sequestration of the background eucalyptus forest is 2.97t per ha.

2. By relieving the children from their daily chores of fetching water and fuelwood, it gives them the opportunity to attend school (Goal 2).
3. As pregnant women do not have to carry water and fuelwood, which may result in miscarriage, and they receive more nutritious meals, maternal health is improved. This in turn reduces child mortality (Goal 4).
4. Environmental sustainability is ensured through provision of safe drinking water, reclamation of degraded land, greening of the environment, and carbon sequestration (Goal 7).
5. Should industrialized countries decide to pay carbon rent to the desert-dwellers in return for greening their habitat, one aspect of global partnership for development may be achieved. This also provides employment opportunities for many desert-dwellers (Goal 8).

6 Food for Thought

1. Contrary to common wisdom, flood is not a proverbial curse but a blessing in disguise for the desert-dwellers. Therefore, the FWS systems, particularly for ARG, should be designed and constructed for the largest possible flood peaks. Ancient water marks on the river walls or trees are good indicators of the maximum expected flood level. The true value of water in deserts makes over-designed systems economically viable.

2. Although we employ rubber-tyre front-end loaders and tracked bulldozers for FWS system construction in sand and firm soils, respectively, each horse power of the earth-moving machinery may be replaced with an able-bodied labourer on an 8-h shift. Thus, 160 unemployed labourers can replace a 160hp bulldozer and install 4ha of FWS system per day for themselves, and make a decent living in most deserts. Shovels, picks, wheel barrows and buckets, the required tools for construction, are available almost anywhere. A piece of transparent hose, filled with water and some colouring agent, replaces the very expensive type at engineering level. In this regard, on average each ha of an ARG system in the Gareh Bygone Plain supplies irrigation water for 0.5ha. Moreover, 5.5ha of irrigated land in the same plain provide a livelihood for a family of 7.64, of owner-operators (farmers who own and work the farm fields; however, as they need help during certain periods of the growing season, they hire labourers for 0.38 part of a year, about 138 days per annum). Therefore, it is safe to assume that the livelihood of a frugal desert-dweller can be gained from 2ha of FWS and 0.5ha of irrigated field under our conditions. *It is better to give small wages instead of alms to the idle masses to make the deserts bloom!* Generous governments always come to the aid of drought-stricken people only after the famine materializes. *Why not prevent famine in the first place by channelling the relief funds to desertification control through floodwater harvesting projects?*
3. The politically expedient and ecologically disastrous sedentarization of transhumant nomads in the Gareh Bygone Plain in the 1930s has generated in their offspring a hostile attitude towards public servants. Moreover, abolishment of the old system, in which an authoritarian chieftain managed a tribe, has left a group of unruly people who do not believe in the sustainable management of marginal drylands. It appears that the nomad mentality cannot accept the rule of law legislated by urbanites. It is unfortunate in this age of enlightenment to resort to force for the benefit of the majority. The absence of a democratic tribal chieftain is therefore badly missed! We hope to make inroads here by proselytizing the younger generation into forming a cooperative together with the research scientists and technicians. We expect that the concrete outcomes of this active research will serve practitioners and policy makers working on water management issues in diverse settings.
4. Assuming a minimum of scientific truth in the reasoning that the mean annual precipitation will decrease in dry areas due to global warming and lead to inevitable famine, it is prudent to prepare for it through the planning and implementation of water harvesting structures in arid and semi-arid zones. Moreover, the rise in sea level due to melting of glaciers may be mitigated, even compensated, by the large-scale floodwater harvesting in the world's deserts.
5. Development and implementation of a water-based economy is essential if the well-being of desert-dwellers is genuinely intended. A hydro-agro-silvo-pastoral economic plan is therefore suggested. It should be emphasized that many factors determine the priority of the last three enterprises. For instance, if the land is not suitable for growing food crops but capable of becoming pasture, a hydro-silvo-pastoral economic programme should be planned and implemented.

We even envision a time when optimization of water use may dictate its sale to the highest bidder. In this case, we forgo growing crops or fruit trees and keep livestock beyond our immediate needs. Floodwater harvesting – particularly for the artificial recharge of groundwater – is technically practicable, environmentally sound, economically viable, and in most cases, socially acceptable and politically prudent.

References

- Anon. 2005. Experts warn ecosystem changes threaten development. *A World of Science*, Vol. 3, No. 3, pp. 8–9.
- Glantz, M. H. 1994. *Drought follows the plow*. Cambridge, UK, Cambridge University Press.
- Kalisz, P. J. and Stone, E. L. 1984. Soil mixing by scarab beetles and pocket gophers in north-central Florida. *Soil Science Society of America Journal*, Vol. 48, pp. 169–172.
- Kowsar, S. A. 1991. Floodwater spreading for desertification control: An integrated approach. *Desertification Control Bulletin*, United Nations Environment Programme, Vol. 19, pp. 3–18.
- . 2005. Abkhandari (aquifer management): A green path to the sustainable development of marginal drylands. *Journal of Mountain Science*, Vol. 2, pp. 233–243.
- Mabbutt, J. A. 1987. Implementation of the Plan of Action to Combat Desertification: Progress since UNCDD. *Land Use Policy*, Vol. 4, pp. 371–388.
- Mohammadnia, M. and Kowsar, S. A. 2003. Clay translocation in the artificial recharge of a groundwater system in the southern Zagros Mountain, Iran. *Mountain Research and Development*, Vol. 23, pp. 50–55.
- Newman, J. C. 1963. Waterspreading on marginal arable areas. *Journal of Soil Conservation, New South Wales*, Vol. 19, pp. 49–58.
- Phillips, J. R. H. 1957. Level-silled bank outlets. *Journal of Soil Conservation, New South Wales*, Vol. 13, No. 2, pp. 15.
- Quilty, J. A. 1972a. Soil conservation structures for marginal arable areas-Gap absorption and gap spreader banks. *Journal of Soil Conservation, New South Wales*, Vol. 2, pp. 116–130.
- . 1972b. Soil conservation structures for marginal arable areas: Diversion spreader banks and tank drains. *Journal of Soil Conservation, New South Wales*, Vol. 28, pp. 157–168.
- Wolman, M. G. 1985. Soil erosion and crop productivity: A worldwide perspective. *Soil Erosion and Crop Productivity*. R. F. Follett and B. A. Stewart (eds.). pp. 9–21.